LECTURE 3

DEEP LEARNING FOR GEOSPATIAL PREDICTION

Geospatial Representation Learning

Learning Outcomes

Lecture

- Explain core deep learning concepts for geospatial prediction tasks.
- Describe how CNNs, transformers, and sequence models are applied to satellite imagery, maps, and time series.
- Distinguish classification, regression, segmentation, retrieval, and forecasting tasks in geospatial AI.
- Evaluate model performance using suitable metrics and validation strategies for spatial data.

Lab

- Implement a basic deep learning workflow for a geospatial prediction task in Python.
- Prepare training, validation, and test splits while considering spatial dependence.
- Train and evaluate a selected model using appropriate loss functions and metrics.
- Interpret prediction results and identify common sources of error or bias.

PRACTICAL 3

TRAINING A GEOSPATIAL PREDICTION MODEL

Geospatial Representation Learning

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